Vectors

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Exercise 0.1 (Store a data point as one object)**.** A house has a floor area of 120 square meters, 3 bedrooms, and an age of 15 years.

1. Write these three numbers as a single vertical list $\underset{\sim}{x}$, in the order given, and call its entries x_1, x_2 and x_3. What is x_2?
2. Does the list $(3, 120, 15)$ describe the same house?

Solution

Solution 0.1.

1. Stacking the three numbers in order gives

$$\underset{\sim}{x} = \begin{bmatrix} 120 \\ 3 \\ 15 \end{bmatrix},$$

   so $x_2 = 3$, the number of bedrooms.

2. No. The list $(3, 120, 15)$ would describe a house with 3 square meters of floor area and 120 bedrooms. An entry's position says what it measures, so the order of the entries matters.

Definition 0.1 (Column vector)**.** A **column vector** [1] of length p is an ordered list of p real numbers, written vertically as a $p \times 1$ array:

$$\underset{\sim}{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix}$$

Remark (Vectors are columns by default)

Remark 0.1 (Vectors are columns by default). In these notes, as in most statistics textbooks, a vector is a column vector unless stated otherwise. A vector written inline as a list in parentheses is still a column vector: for example, $\underset{\sim}{x} = (2, -1, 5)$ means

$$\underset{\sim}{x} = \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix}.$$

Remark (Why the vector mark sits under the symbol)

Remark 0.2 (Why the vector mark sits under the symbol). In these notes, the mark that shows a symbol is a vector is a tilde under the letter, as in $\underset{\sim}{x}$. Many books instead use a bold letter or an arrow over the letter. We chose a mark underneath because bold letters are hard to write by hand on a board, and because the space above a letter is already used by the hat that marks an estimated value, as in $\widehat{\beta}$. The notation choices page gives the full reasons and an example of an estimated vector$\widetilde{\text{r}}$.

Exercise 0.2 (Reading one entry of a column vector of temperatures)**.** A week of daily readings is often stored as a column vector, with the reading for day i in entry i. The noon temperatures on the Celsius scale on the first four days of a week are

$$\underset{\sim}{x} = \begin{bmatrix} 12 \\ 15 \\ 9 \\ 11 \end{bmatrix}.$$

Find x_3.

Hint

In Definition 0.1, x_i is the entry in position i, counting from the top.

Solution

Solution 0.2. $x_3 = 9$, the third entry from the top, so the noon temperature on day 3 was 9 on the Celsius scale.

Definition 0.2 (The set $\mathbb{R}^p$)**.** For a positive integer p, $\mathbb{R}^p$ is the set of all column vectors of length p whose entries are real numbers.

Remark (Reading $\underset{\sim}{x} \in \mathbb{R}^p$)

Remark 0.3 (Reading $\underset{\sim}{x} \in \mathbb{R}^p$). Writing $\underset{\sim}{x} \in \mathbb{R}^p$ says, in one symbol, that $\underset{\sim}{x}$ is a column vector with p real entries. For example, $(1, -2, 0.5) \in \mathbb{R}^3$. The vector $(1, 2)$ is not in $\mathbb{R}^3$, because it has only 2 entries; it is in $\mathbb{R}^2$. In Exercise 0.1, $\underset{\sim}{x} \in \mathbb{R}^3$.

Exercise 0.3 (Testing whether a vector with entries $\sqrt{2}$ and π lies in $\mathbb{R}^3$)**.** Entries such as $\sqrt{2}$ and π have no exact decimal form, but they are still real numbers. Let

$$\underset{\sim}{v} = \begin{bmatrix} \sqrt{2} \\ \pi \\ -1 \end{bmatrix}.$$

Is $\underset{\sim}{v}$ in $\mathbb{R}^3$?

Hint

Definition 0.2 has two conditions: the number of entries of the column vector, and the kind of number in each entry.

Solution

Solution 0.3. Yes. $\underset{\sim}{v}$ is a column vector with 3 entries, and each of $\sqrt{2}$, π and -1 is a real number, so $\underset{\sim}{v} \in \mathbb{R}^3$ by Definition 0.2.

Definition 0.3 (Scalar)**.** In linear algebra, a **scalar** [2] is a single real number, as opposed to a vector or a matrix, which holds several numbers.

Example 0.1 (Scalars and a vector)**.** The numbers 5 and -0.2 are scalars. The column vector $(5, -0.2)$ is not a scalar: it has 2 entries, so it is in $\mathbb{R}^2$ (Definition 0.2), and each of its 2 entries is a scalar.

Exercise 0.4 (Picking out the scalars among class summaries)**.** A summary of a class of 30 students can report single numbers or whole lists of numbers. Consider these three quantities:

- the height of the tallest student in the class,
- the list of every student's height,
- the number of students in the class.

Which of these three quantities are scalars?

Hint

By Definition 0.3, a scalar is one real number. Count the numbers in each quantity.

Solution

Solution 0.4. The tallest student's height and the number of students are scalars: each is one real number. The list of heights holds 30 numbers, so it is a column vector in $\mathbb{R}^{30}$, not a scalar.

Definition 0.4 (Real-valued function, scalar-valued function)**.** A function $f : A \to \mathbb{R}$, whose values are numbers, is **real-valued** (also called **scalar-valued**, since its values are scalars).

Definition 0.5 (Vector-valued function)**.** A function $f : A \to \mathbb{R}^m$, whose values are vectors (Definition 0.2), is **vector-valued**.

Example 0.2 (A real-valued and a vector-valued function)**.** On $\mathbb{R}^2$, $f(\underset{\sim}{x}) = x_1 + x_2$ is real-valued and $g(\underset{\sim}{x}) = \begin{bmatrix} x_1 + x_2 \\ x_1 - x_2 \end{bmatrix}$ is vector-valued:

$$f\left(\begin{bmatrix} 3 \\ 4 \end{bmatrix}\right) = 3 + 4 = 7, \qquad g\left(\begin{bmatrix} 3 \\ 4 \end{bmatrix}\right) = \begin{bmatrix} 3 + 4 \\ 3 - 4 \end{bmatrix} = \begin{bmatrix} 7 \\ -1 \end{bmatrix}$$

Exercise 0.5 (Testing whether a price prediction is real-valued)**.** A model predicts a house's price from three of its features, stored in $\underset{\sim}{x} \in \mathbb{R}^3$, with the function

$$f(\underset{\sim}{x}) = 2x_1 + 5x_2 - x_3.$$

Is f real-valued?

Hint

Evaluate f at one point, and count the numbers in the result.

Solution

Solution 0.5. Yes. For every $\underset{\sim}{x} \in \mathbb{R}^3$, $2x_1 + 5x_2 - x_3$ is one real number, so $f : \mathbb{R}^3 \to \mathbb{R}$ is real-valued. For example, $f\left(\begin{bmatrix}1\\1\\1\end{bmatrix}\right) = 2 + 5 - 1 = 6$.

Exercise 0.6 (Evaluating a vector-valued function at one input)**.** A vector-valued function returns several numbers for one input. Let $g : \mathbb{R}^2 \to \mathbb{R}^3$ be

$$g(\underset{\sim}{x}) = \begin{bmatrix} x_1 + x_2 \\ x_1 x_2 \\ x_1 - x_2 \end{bmatrix}.$$

Find $g\left(\begin{bmatrix}2\\3\end{bmatrix}\right)$.

Hint

Substitute $x_1 = 2$ and $x_2 = 3$ into each entry, as in Example 0.2.

Solution

Solution 0.6. Substituting $x_1 = 2$ and $x_2 = 3$,

$$g\left(\begin{bmatrix}2\\3\end{bmatrix}\right) = \begin{bmatrix} 2+3 \\ 2 \cdot 3 \\ 2-3 \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \\ -1 \end{bmatrix},$$

a column vector in $\mathbb{R}^3$.

Definition 0.6 (Transpose)**.** The **transpose** of a column vector $\underset{\sim}{x}$ is the row vector with the same sequence of entries, written horizontally:

$$\underset{\sim}{x}^{\top} \equiv \underset{\sim}{x}' \equiv [x_1,\ x_2,\ \ldots,\ x_p]$$

Example 0.3 (Transposing a column vector)**.** The transpose of the column vector with entries 2, −1, 5 is the row vector with the same entries:

$$\begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix}^{\top} = [2,\ -1,\ 5].$$

Exercise 0.7 (Writing one observation as a row vector)**.** A data table stores each observation on its own line, so an observation held as a column vector is written in the table as its transpose. Let

$$\underset{\sim}{x} = \begin{bmatrix} 4 \\ -1 \\ 0 \end{bmatrix}.$$

Find $\underset{\sim}{x}^{\top}$.

Hint

By Definition 0.6, keep the entries in the same order and write them horizontally.

Solution

Solution 0.7. By Definition 0.6,

$$\underset{\sim}{x}^{\top} = [4,\ -1,\ 0].$$

Definition 0.7 (Vector addition)**.** The **sum** of two column vectors $\underset{\sim}{x}$ and y of the same length p is the column vector $\underset{\sim}{x} + \underset{\sim}{y}$ of length p obtained by adding entry by entry:

$$(\underset{\sim}{x} + \underset{\sim}{y})_i \overset{\text{def}}{=} x_i + y_i, \quad i = 1, \ldots, p$$

Example 0.4 (Adding two vectors)**.**

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1+4 \\ 2+5 \\ 3+6 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \\ 9 \end{bmatrix}$$

Exercise 0.8 (Adding two months of sales for three items)**.** A shop records how many units of each of three items it sells in a month, as a column vector in $\mathbb{R}^3$. In two months it sells

$$\underset{\sim}{x} = \begin{bmatrix} 12 \\ 5 \\ 8 \end{bmatrix} \quad \text{and} \quad \underset{\sim}{y} = \begin{bmatrix} 10 \\ 7 \\ 3 \end{bmatrix}$$

units.
Find $\underset{\sim}{x} + \underset{\sim}{y}$, the units of each item sold over both months.

Hint

Use Definition 0.7, as in Example 0.4.

Solution

Solution 0.8. By Definition 0.7,

$$\underset{\sim}{x} + \underset{\sim}{y} = \begin{bmatrix} 12+10 \\ 5+7 \\ 8+3 \end{bmatrix} = \begin{bmatrix} 22 \\ 12 \\ 11 \end{bmatrix}.$$

Definition 0.8 (Dot product)**.** For any two real-valued vectors $\underset{\sim}{x} = (x_1, \ldots, x_p)$ and $\underset{\sim}{y} = (y_1, \ldots, y_p)$ of the same length p, the **dot product** of $\underset{\sim}{x}$ and $\underset{\sim}{y}$ is:

$$\underset{\sim}{x} \cdot \underset{\sim}{y} = \underset{\sim}{x}^\top \underset{\sim}{y} \overset{\text{def}}{=} \sum_{i=1}^{p} x_i y_i \tag{1}$$

See also the definitions in:

- Dobson and Barnett [3], Section 1.3 (equation 1.1, page 7)
- Kaplan [4], chapter on vectors
- wikipedia

The dot product has a different generalization for two matrices; see wikipedia for more.

Example 0.5 (A dot product)**.** For $\underset{\sim}{x} = (1, 2, 3)$ and $\underset{\sim}{y} = (4, 5, 6)$:

$$\begin{aligned} \underset{\sim}{x} \cdot \underset{\sim}{y} &= 1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6 && \text{(definition of the dot product)} \\ &= 4 + 10 + 18 && \text{(multiply)} \\ &= 32 && \text{(add)} \end{aligned}$$

Definition 0.9 (Linear combination)**.** A **linear combination** of the numbers $a_1, \ldots, a_k$ with coefficients $c_1, \ldots, c_k$ is the sum

$$c_1 a_1 + \cdots + c_k a_k = \sum_{i=1}^{k} c_i a_i$$

A linear combination of vectors $\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k$ of the same length is defined entry by entry: entry j of $\sum_{i=1}^{k} c_i \underset{\sim}{v}_i$ is the linear combination of the jth entries of $\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k$ with the same coefficients.

Definition 0.10 (Coefficients of a linear combination)**.** The numbers $c_1, \ldots, c_k$ in a linear combination (Definition 0.9) $\sum_{i=1}^{k} c_i a_i$ are its **coefficients**.

Example 0.6 (Coefficients of a linear combination of numbers)**.** The linear combination of the numbers 4 and 5 with coefficients 2 and 3 is $2 \cdot 4 + 3 \cdot 5 = 8 + 15 = 23$.

Definition 0.11 (Weighted sum)**.** A **weighted sum** of the numbers $a_1, \ldots, a_k$ is a linear combination of them, with coefficients $w_1, \ldots, w_k$:

$$w_1 a_1 + \cdots + w_k a_k = \sum_{i=1}^{k} w_i a_i$$

Multiply each a_i by its coefficient w_i, then add.

Definition 0.12 (Weight)**.** The coefficients $w_1, \ldots, w_k$ of a weighted sum $\sum_{i=1}^{k} w_i a_i$ are called its **weights**. The weight w_i says how much a_i counts.

Example 0.7 (The total price of a shopping basket)**.** A basket holds 3 items that cost \$2 each and 2 items that cost \$5 each. The total price is the weighted sum of the prices 2 and 5 with weights 3 and 2, the count of items at each price:

$$\begin{aligned} 3 \cdot 2 + 2 \cdot 5 &= 6 + 10 && \text{(multiply each price by its weight)} \\ &= 16 && \text{(add)} \end{aligned}$$

Definition 0.13 (Weighted average, weighted mean)**.** A **weighted average** (or **weighted mean**) of the numbers $a_1, \dots, a_k$ is a weighted sum of them whose weights $w_1, \dots, w_k$ are non-negative and add to 1:

$$\sum_{i=1}^{k} w_i a_i \quad \text{with} \quad w_i \geq 0 \text{ for each } i \quad \text{and} \quad \sum_{i=1}^{k} w_i = 1$$

An a_i with a larger weight counts for more.

Example 0.8 (A course grade)**.** A course grade counts the midterm for 1/4 and the final for 3/4. A student scores 70 on the midterm and 90 on the final. The weights 1/4 and 3/4 are non-negative and add to 1, so the grade is a weighted average of 70 and 90:

$$\begin{aligned} \frac{1}{4} \cdot 70 + \frac{3}{4} \cdot 90 &= 17.5 + 67.5 && \text{(multiply each score by its weight)} \\ &= 85 && \text{(add)} \end{aligned}$$

The grade 85 lies between 70 and 90, and nearer to 90, which has the larger weight.

Example 0.9 (A linear combination of two vectors)**.** The linear combination of $(1, 0)^\top$ and $(0, 1)^\top$ with coefficients 2 and 3 is $(2 \cdot 1 + 3 \cdot 0,\ 2 \cdot 0 + 3 \cdot 1)^\top = (2, 3)^\top$.

Remark (The dot product is a linear combination)

Remark 0.4 (The dot product is a linear combination). The dot product $\underset{\sim}{x} \cdot \underset{\sim}{y}$ is a linear combination (Definition 0.9) of the entries of y, with coefficients $x_1, \dots, x_p$. In Example 0.5, $\underset{\sim}{x} \cdot \underset{\sim}{y} = 1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6$ is the linear combination of 4, 5, and 6 with coefficients 1, 2, and 3.
The dot product is also the standard *inner product* on $\mathbb{R}^p$; "inner product" is the general notion (Definition 5 in Inner Products and Orthogonality, later in these notes), of which the dot product is one example.

Theorem 0.1 (Dot product is symmetric)**.** *The dot product is symmetric:*

$$\underset{\sim}{x} \cdot \underset{\sim}{y} = \underset{\sim}{y} \cdot \underset{\sim}{x}$$

Proof

Proof.

$$\begin{aligned} \underset{\sim}{x} \cdot \underset{\sim}{y} &= \sum_{i=1}^{p} x_i y_i && \text{(definition of the dot product)} \\ &= \sum_{i=1}^{p} y_i x_i && \text{(products of numbers are symmetric)} \\ &= \underset{\sim}{y} \cdot \underset{\sim}{x} && \text{(definition of the dot product)} \end{aligned}$$

□

Special vectors

Definition 0.1 (Zero vector)**.** The **zero vector** $\underset{\sim}{0}$ of length p has all entries equal to zero:

$$\underset{\sim}{0} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Remark (The zero vector is the additive identity)

Remark 0.1 (The zero vector is the additive identity). Adding the zero vector to a vector leaves it unchanged (Definition 0.7): $\underset{\sim}{x} + \underset{\sim}{0} = \underset{\sim}{x}$ for any vector $\underset{\sim}{x}$ of the same length, so $\underset{\sim}{0}$ is the identity element for vector addition. For example,

$$\begin{aligned} (2, -1) + (0, 0) &= (2 + 0, -1 + 0) \\ &= (2, -1). \end{aligned}$$

Exercise 0.1 (Picking the zero vector with three entries from three candidates)**.** A zero vector has a number of entries as well as zero entries, so $\underset{\sim}{0} \in \mathbb{R}^2$ and $\underset{\sim}{0} \in \mathbb{R}^3$ are different vectors. Consider three candidates:

$$\underset{\sim}{a} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \qquad \underset{\sim}{b} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \qquad \underset{\sim}{c} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

Which of the three candidates is the zero vector with 3 entries?

Hint

Definition 0.1 has two conditions: 3 entries, and 0 in every entry.

Solution

Solution 0.1. Only $\underset{\sim}{b}$. $\underset{\sim}{a}$ has only 2 entries, so it is the zero vector in $\mathbb{R}^2$, and $\underset{\sim}{c}$ has a nonzero second entry.

Definition 0.2 (Mean, average)**.** The **mean** [5] (or **average**) of n numbers $y_1, \ldots, y_n$ is their sum divided by how many there are:

$$\bar{y} \stackrel{\text{def}}{=} \frac{1}{n} \sum_{i=1}^{n} y_i$$

Example 0.1 (The mean of three numbers)**.** For $y_1 = 2$, $y_2 = -1$ and $y_3 = 5$:

$$\begin{aligned} \bar{y} &= \frac{1}{3}(2 + (-1) + 5) && \text{(definition of the mean, with } n = 3) \\ &= \frac{1}{3} \cdot 6 && \text{(add)} \\ &= 2 && \text{(multiply)} \end{aligned}$$

Remark (The sample mean estimates the population mean)

Remark 0.2 (The sample mean estimates the population mean). When the numbers are observed data values, $\bar{y}$ is also called the sample mean, an estimator of the population mean.

Exercise 0.2 (Finding the mean age of four siblings)**.** Four siblings are 3, 7, 10 and 12 years old.
Find the mean of their ages.

Hint

Use Definition 0.2 with $n = 4$, as in Example 0.1.

Solution

Solution 0.2. By Definition 0.2 with $n = 4$,

$$\bar{y} = \frac{1}{4}(3 + 7 + 10 + 12) = \frac{32}{4} = 8,$$

so the mean age is 8 years.

Definition 0.3 (Ones vector)**.** The **ones vector** $\underset{\sim}{1}$ of length p has all entries equal to one:

$$\underset{\sim}{1} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

Exercise 0.3 (Adding one to every entry with the ones vector)**.** Adding the ones vector to a vector raises every entry by 1, for example to shift a list of counts that start at 0 so that they start at 1. Let

$$\underset{\sim}{x} = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}.$$

Find $\underset{\sim}{x} + \underset{\sim}{1}$, where $\underset{\sim}{1}$ has 3 entries.

Hint

Write out $\underset{\sim}{1}$ with Definition 0.3, then use Definition 0.7.

Solution

Solution 0.3. By Definition 0.3, $\underset{\sim}{1} = (1, 1, 1)^\top$, so by Definition 0.7

$$\underset{\sim}{x} + \underset{\sim}{1} = \begin{bmatrix} 2+1 \\ 0+1 \\ -1+1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}.$$

Definition 0.4 (Sum of the elements of a vector)**.** The **sum of the elements** of a vector $\underset{\sim}{x} \in \mathbb{R}^p$ is the sum of its entries:

$$s_{\underset{\sim}{x}} \overset{\text{def}}{=} \sum_{i=1}^{p} x_i.$$

In words, $s_{\underset{\sim}{x}}$ is the entries of $\underset{\sim}{x}$ added together.

Example 0.2 (The sum of the elements of a vector with three entries)**.** For $\underset{\sim}{x} = (2, -1, 5)$,

$$\begin{aligned} s_{\underset{\sim}{x}} &= 2 + (-1) + 5 && \text{(definition of the sum of the elements, with } p = 3\text{)} \\ &= 6. && \text{(add)} \end{aligned}$$

Exercise 0.4 (Totalling the units sold across three items)**.** A shop sells 22, 12 and 11 units of three items over two months, stored as

$$\underset{\sim}{x} = \begin{bmatrix} 22 \\ 12 \\ 11 \end{bmatrix}.$$

Find $s_{\underset{\sim}{x}}$, the sum of the elements of $\underset{\sim}{x}$.

Hint

Use Definition 0.4 with $p = 3$, as in Example 0.2.

Solution

Solution 0.4. By Definition 0.4,

$$s_{\underset{\sim}{x}} = 22 + 12 + 11 = 45,$$

so the shop sold 45 units in all.

Theorem 0.1 (The ones vector sums the entries)**.** *For every $\underset{\sim}{x} \in \mathbb{R}^p$, the sum of the elements of $\underset{\sim}{x}$ is the dot product of the ones vector with $\underset{\sim}{x}$:*

$$s_{\underset{\sim}{x}} = \underset{\sim}{1} \cdot \underset{\sim}{x}.$$

Proof

Proof.

$$\begin{aligned} \underset{\sim}{1} \cdot \underset{\sim}{x} &= \sum_{i=1}^{p} 1 \cdot x_i && \text{(Definition 0.8, with every entry of } \underset{\sim}{1} \text{ equal to 1)} \\ &= \sum_{i=1}^{p} x_i && \text{(multiply each entry by 1)} \\ &= s_{\underset{\sim}{x}}. && \text{(Definition 0.4)} \end{aligned}$$

□

Example 0.3 (The ones vector applied to a vector with three entries)**.** For $\underset{\sim}{x} = (2, -1, 5)$ and the ones vector $\underset{\sim}{1}$ with three entries,

$$\begin{aligned} \underset{\sim}{1} \cdot \underset{\sim}{x} &= 1 \cdot 2 + 1 \cdot (-1) + 1 \cdot 5 && \text{(Definition 0.8)} \\ &= 6, && \text{(multiply and add)} \end{aligned}$$

which is $s_{\underset{\sim}{x}}$, as in Example 0.2. Dividing by $p = 3$ gives the mean of the entries, $6/3 = 2$.

Exercise 0.5 (Summing entries with a dot product against the ones vector)**.** Software that computes dot products quickly can total the entries of a vector through Theorem 0.1. Let

$$\underset{\sim}{x} = \begin{bmatrix} 4 \\ -2 \\ 7 \\ 1 \end{bmatrix},$$

and let $\underset{\sim}{1}$ be the ones vector with 4 entries.
Find $\underset{\sim}{1} \cdot \underset{\sim}{x}$.

Hint

By Theorem 0.1, $\underset{\sim}{1} \cdot \underset{\sim}{x}$ is the sum of the elements of $\underset{\sim}{x}$.

Solution

Solution 0.5. By Theorem 0.1,

$$\underset{\sim}{1} \cdot \underset{\sim}{x} = s_{\underset{\sim}{x}} = 4 + (-2) + 7 + 1 = 10.$$

Definition 0.5 (Indicator vector, standard basis vector)**.** The j-th **indicator vector** (or **standard basis vector** [6]) $\underset{\sim}{e}_j$ of length p has a 1 in position j and 0s elsewhere:

$$(\underset{\sim}{e}_j)_i = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \qquad \underset{\sim}{e}_j = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \leftarrow \text{position } j$$

Remark (Indicator vectors build every vector)

Remark 0.3 (Indicator vectors build every vector). For $p = 3$, the indicator vectors are $\underset{\sim}{e}_1 = (1, 0, 0)$, $\underset{\sim}{e}_2 = (0, 1, 0)$, and $\underset{\sim}{e}_3 = (0, 0, 1)$. Every vector of length 3 is a weighted sum $c_1\underset{\sim}{e}_1 + c_2\underset{\sim}{e}_2 + c_3\underset{\sim}{e}_3$ of them, with weights equal to its entries; for example, $(2, -1, 5) = 2\underset{\sim}{e}_1 - 1\underset{\sim}{e}_2 + 5\underset{\sim}{e}_3$. No other weights work: entry j of $c_1\underset{\sim}{e}_1 + c_2\underset{\sim}{e}_2 + c_3\underset{\sim}{e}_3$ is c_j, so each weight c_j must equal entry j of the vector.

Exercise 0.6 (Encoding the second of four categories as an indicator vector)**.** One-hot encoding stores a value that falls in one of p categories as the indicator vector of its category. A survey answer falls in the second of 4 categories.
Find $\underset{\sim}{e}_2$ with 4 entries, the one-hot encoding of that answer.

Hint

By Definition 0.5, $\underset{\sim}{e}_2$ has a 1 in position 2 and 0s elsewhere.

Solution

Solution 0.6. By Definition 0.5 with $j = 2$ and $p = 4$,

$$\underset{\sim}{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}.$$

Theorem 0.2 (Indicator vectors select entries)**.** *For any vector* $\underset{\sim}{x}$ *of length* p *and any* $j \in \{1, \ldots, p\}$*:*

$$\underset{\sim}{e}_j \cdot \underset{\sim}{x} = x_j$$

Proof

Proof. Writing the dot product componentwise:

$$
\begin{aligned}
\underset{\sim}{e}_j \cdot \underset{\sim}{x} &= \sum_{i=1}^{p} (\underset{\sim}{e}_j)_i \, x_i && \text{(definition of the dot product)} \\
&= \sum_{i=1}^{p} \begin{cases} 1 \cdot x_i & \text{if } i = j \\ 0 \cdot x_i & \text{if } i \neq j \end{cases} && \text{(definition of } \underset{\sim}{e}_j\text{)} \\
&= x_j && \text{(only the } i = j \text{ term is nonzero)}
\end{aligned}
$$

□

Exercise 0.7 (Selecting one entry with an indicator vector)**.** Theorem 0.2 turns reading one entry of a vector into a dot product. Let

$$
\underset{\sim}{x} = \begin{bmatrix} 5 \\ -1 \\ 8 \\ 2 \end{bmatrix}.
$$

Find $\underset{\sim}{e}_3 \cdot \underset{\sim}{x}$, where $\underset{\sim}{e}_3$ is the indicator vector with 4 entries.

Hint

Use Theorem 0.2 with $j = 3$.

Solution

Solution 0.7. By Theorem 0.2 with $j = 3$, $\underset{\sim}{e}_3 \cdot \underset{\sim}{x} = x_3 = 8$. Computing the dot product directly gives the same answer:

$$
\underset{\sim}{e}_3 \cdot \underset{\sim}{x} = 0 \cdot 5 + 0 \cdot (-1) + 1 \cdot 8 + 0 \cdot 2 = 8.
$$

Theorem 0.3 (A vector is a linear combination of indicator vectors)**.** *Every vector $\underset{\sim}{x} \in \mathbb{R}^p$ is the linear combination (Definition 0.9) of the indicator vectors $\underset{\sim}{e}_1, \ldots, \underset{\sim}{e}_p$ (Definition 0.5) with coefficients $x_1, \ldots, x_p$:*

$$
\underset{\sim}{x} = \sum_{j=1}^{p} x_j \underset{\sim}{e}_j
$$

Proof

Proof. Compare the two sides entry by entry. For each $i \in \{1, \ldots, p\}$:

$$
\begin{aligned}
\left(\sum_{j=1}^{p} x_j \underset{\sim}{e}_j\right)_i &= \sum_{j=1}^{p} x_j (\underset{\sim}{e}_j)_i && \text{(Definition 0.9, entry by entry)} \\
&= \sum_{j=1}^{p} \begin{cases} x_j \cdot 1 & \text{if } j = i \\ x_j \cdot 0 & \text{if } j \neq i \end{cases} && \text{(Definition 0.5)} \\
&= x_i \cdot 1 + \sum_{j \neq i} x_j \cdot 0 && \text{(split off the } j = i \text{ term)} \\
&= x_i + 0 && \text{(times 1 changes nothing; times 0 gives 0)} \\
&= x_i && \text{(adding 0 changes nothing)}
\end{aligned}
$$

□

Example 0.4 (Expanding a vector in indicator vectors)**.**

$$
\begin{aligned}
\begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix} &= 4 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + (-1) \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \\
&= 4\underset{\sim}{e}_1 - \underset{\sim}{e}_2 + 2\underset{\sim}{e}_3
\end{aligned}
$$

Exercise 0.8 (Writing a vector as a linear combination of indicator vectors)**.** By Theorem 0.3, every vector in $\mathbb{R}^3$ is a linear combination of the indicator vectors $\underset{\sim}{e}_1$, $\underset{\sim}{e}_2$ and $\underset{\sim}{e}_3$. Let

$$
\underset{\sim}{x} = \begin{bmatrix} -3 \\ 0 \\ 6 \end{bmatrix}.
$$

Find the numbers c_1, c_2 and c_3 with $\underset{\sim}{x} = c_1\underset{\sim}{e}_1 + c_2\underset{\sim}{e}_2 + c_3\underset{\sim}{e}_3$.

Hint

Use Theorem 0.3, as in Example 0.4.

Solution

Solution 0.8. By Theorem 0.3, the numbers c_j are the entries of $\underset{\sim}{x}$: $c_1 = -3$, $c_2 = 0$ and $c_3 = 6$. As a check,

$$-3\begin{bmatrix}1\\0\\0\end{bmatrix}+0\begin{bmatrix}0\\1\\0\end{bmatrix}+6\begin{bmatrix}0\\0\\1\end{bmatrix}=\begin{bmatrix}-3\\0\\6\end{bmatrix}.$$

Orthogonality

Definition 0.1 (Orthogonal vectors)**.** Two vectors $\underset{\sim}{x}$ and $\underset{\sim}{y}$ of the same length are **orthogonal** [7] (written $\underset{\sim}{x} \perp \underset{\sim}{y}$) if their dot product (Definition 0.8) is zero:

$$\underset{\sim}{x} \perp \underset{\sim}{y} \iff \underset{\sim}{x} \cdot \underset{\sim}{y} = 0$$

Remark (Orthogonal means perpendicular)

Remark 0.1 (Orthogonal means perpendicular). Orthogonality extends the geometric idea of perpendicular lines to vectors with any number of entries. In the plane, $(1, 2)$ and $(-2, 1)$ are perpendicular, and their dot product is $1 \cdot (-2) + 2 \cdot 1 = 0$. The same test works in $\mathbb{R}^3$, where pictures are harder to draw:

$$\begin{aligned}(1, 1, 0) \cdot (1, -1, 5) &= 1 - 1 + 0\\ &= 0,\end{aligned}$$

so $(1, 1, 0) \perp (1, -1, 5)$.

Example 0.1 (Vectors that are not orthogonal)**.**

$$\begin{aligned}(1, 2) \cdot (1, 1) &= 1 \cdot 1 + 2 \cdot 1\\ &= 3\\ &\neq 0,\end{aligned}$$

so $(1, 2)$ and $(1, 1)$ are not orthogonal.

Exercise 0.1 (Choosing c to make two vectors orthogonal)**.** Principal component analysis describes data along orthogonal vectors, so testing for orthogonality comes up often. For $c \in \mathbb{R}$, let

$$\underset{\sim}{x} = \begin{bmatrix} 2 \\ c \\ 1 \end{bmatrix}, \qquad \underset{\sim}{y} = \begin{bmatrix} 1 \\ 3 \\ -5 \end{bmatrix}.$$

Find the value of c for which $\underset{\sim}{x} \perp \underset{\sim}{y}$.

Hint

By Definition 0.1, solve $\underset{\sim}{x} \cdot \underset{\sim}{y} = 0$ for c, using Definition 0.8.

Solution

Solution 0.1. $c = 1$. By Definition 0.1, $\underset{\sim}{x} \perp \underset{\sim}{y}$ exactly when

$$\begin{aligned} 0 &= \underset{\sim}{x} \cdot \underset{\sim}{y} && \text{(Definition 0.1)} \\ &= 2 \cdot 1 + c \cdot 3 + 1 \cdot (-5) && \text{(Definition 0.8)} \\ &= 3c - 3, && \text{(multiply and collect terms)} \end{aligned}$$

which holds only for $c = 1$.

Definition 0.2 (Euclidean norm (length, L_2 norm))**.** The **Euclidean norm** [8] (or **length**, also called the L_2 **norm** and written $\|\underset{\sim}{x}\|_2$) of a vector $\underset{\sim}{x}$ of length p is the square root of the sum of the squares of its entries:

$$\|\underset{\sim}{x}\| \stackrel{\text{def}}{=} \sqrt{\sum_{i=1}^{p} x_i^2} \tag{2}$$

Example 0.2 (The length of a vector)**.** For $\underset{\sim}{x} = (3, 4)$:

$$\begin{aligned} \|\underset{\sim}{x}\| &= \sqrt{3^2 + 4^2} && \text{(definition of the norm)} \\ &= \sqrt{25} && \text{(square and add)} \\ &= 5 \end{aligned}$$

The vector $(0.6, 0.8)$ has norm $\sqrt{0.36 + 0.64} = 1$.

Theorem 0.1 (Euclidean norm from the dot product)**.** *The Euclidean norm of a vector $\underset{\sim}{x}$ is the square root of the dot product of $\underset{\sim}{x}$ with itself:*

$$\|\underset{\sim}{x}\| = \sqrt{\underset{\sim}{x} \cdot \underset{\sim}{x}} \tag{3}$$

Proof

Proof.

$$\begin{aligned}
\|\underset{\sim}{x}\| &= \sqrt{\sum_{i=1}^{p} x_i^2} && \text{(Equation 2)} \\
&= \sqrt{\sum_{i=1}^{p} x_i x_i} && (x_i^2 = x_i x_i) \\
&= \sqrt{\underset{\sim}{x} \cdot \underset{\sim}{x}} && \text{(Equation 1, with } \underset{\sim}{y} = \underset{\sim}{x})
\end{aligned}$$

□

Example 0.3 (The length of a vector from its dot product)**.** For $\underset{\sim}{x} = (3, 4)$ from Example 0.2:

$$\begin{aligned}
\underset{\sim}{x} \cdot \underset{\sim}{x} &= 3 \cdot 3 + 4 \cdot 4 && \text{(Equation 1)} \\
&= 9 + 16 && \text{(multiply)} \\
&= 25 && \text{(add)}
\end{aligned}$$

so Theorem 0.1 gives $\|\underset{\sim}{x}\| = \sqrt{25} = 5$, the same length that Example 0.2 found from Equation 2.

Exercise 0.2 (Finding the norm of a vector with four entries)**.** The Euclidean norm works the same way in $\mathbb{R}^4$ as in the plane, although there is no picture to draw. Let $\underset{\sim}{x}$ be the vector

$$\underset{\sim}{x} = \begin{bmatrix} 1 \\ 2 \\ 2 \\ 4 \end{bmatrix}.$$

Find $\|\underset{\sim}{x}\|$.

Hint

Use Definition 0.2 with $p = 4$, as in Example 0.2.

Solution

Solution 0.2. By Definition 0.2,

$$\|\underset{\sim}{x}\| = \sqrt{1^2 + 2^2 + 2^2 + 4^2} = \sqrt{25} = 5.$$

Definition 0.3 (Unit vector)**.** A **unit vector** [9] is a vector whose Euclidean norm (Definition 0.2) is 1. A unit vector is said to have *unit length*.

Example 0.4 (A unit vector and a vector that is not one)**.** For $\underset{\sim}{x} = (0.6, 0.8)$:

$$\begin{aligned} \|\underset{\sim}{x}\| &= \sqrt{0.6^2 + 0.8^2} && \text{(definition of the norm)} \\ &= \sqrt{0.36 + 0.64} && \text{(square)} \\ &= \sqrt{1} && \text{(add)} \\ &= 1 && \text{(take the square root)} \end{aligned}$$

so $(0.6, 0.8)$ is a unit vector. The vector $(1, 1)$ is not:

$$\begin{aligned} \|(1,1)\| &= \sqrt{1^2 + 1^2} \\ &= \sqrt{2} \\ &\neq 1. \end{aligned}$$

Exercise 0.3 (How far apart are two points?)**.** Let $\underset{\sim}{x} = (1, 2)$ and $\underset{\sim}{y} = (4, 6)$.

1. Compute the vector $\underset{\sim}{x} - \underset{\sim}{y}$.
2. Compute the length of $\underset{\sim}{x} - \underset{\sim}{y}$, using the Euclidean norm.
3. Is the result the same if you compute the length of $\underset{\sim}{y} - \underset{\sim}{x}$ instead?

Solution

Solution 0.3.

1. $$\begin{aligned} \underset{\sim}{x} - \underset{\sim}{y} &= (1 - 4, 2 - 6) \\ &= (-3, -4). \end{aligned}$$

2. $$\begin{aligned} \|\underset{\sim}{x} - \underset{\sim}{y}\| &= \sqrt{(-3)^2 + (-4)^2} \\ &= \sqrt{25} \\ &= 5. \end{aligned}$$

3. $\underset{\sim}{y} - \underset{\sim}{x} = (3, 4)$, and
$$\begin{aligned} \|\underset{\sim}{y} - \underset{\sim}{x}\| &= \sqrt{3^2 + 4^2} \\ &= 5. \end{aligned}$$

Yes, the result is the same.

Definition 0.4 (Euclidean distance)**.** The **Euclidean distance** [10] between two vectors $\underset{\sim}{x}$ and $\underset{\sim}{y}$ of length p is the Euclidean norm of their difference:

$$d(\underset{\sim}{x}, \underset{\sim}{y}) \stackrel{\text{def}}{=} \|\underset{\sim}{x} - \underset{\sim}{y}\| = \sqrt{\sum_{i=1}^{p} (x_i - y_i)^2} \tag{4}$$

Remark (Distance from the origin)

Remark 0.2 (Distance from the origin). The distance from the origin $\underset{\sim}{0}$ to a vector $\underset{\sim}{x}$ is the norm of $\underset{\sim}{x}$:

$$\begin{aligned} d(\underset{\sim}{x}, \underset{\sim}{0}) &= \|\underset{\sim}{x} - \underset{\sim}{0}\| \\ &= \|\underset{\sim}{x}\| \end{aligned}$$

[11, p. 39].
For example, the distance from $\underset{\sim}{0}$ to $(3, 4)$ is $\|(3, 4)\| = 5$, as in Example 0.2.

Exercise 0.4 (Finding the distance between two customers' feature vectors)**.** A nearest-neighbour method compares two customers through the Euclidean distance between their feature vectors. Two customers have feature vectors

$$\underset{\sim}{x} = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}, \qquad \underset{\sim}{y} = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}.$$

Find $d(\underset{\sim}{x}, \underset{\sim}{y})$.

Hint

Use Definition 0.4 with $p = 3$.

Solution

Solution 0.4. By Definition 0.4,

$$\begin{aligned} d(\underset{\sim}{x}, \underset{\sim}{y}) &= \sqrt{(1-3)^2 + (0-1)^2 + (3-1)^2} && \text{(Definition 0.4)} \\ &= \sqrt{4 + 1 + 4} && \text{(subtract and square)} \\ &= 3. && \text{(add and take the square root)} \end{aligned}$$

Definition 0.5 (Orthonormal vectors)**.** A set of vectors $\{\underset{\sim}{x}_1, \underset{\sim}{x}_2, \ldots, \underset{\sim}{x}_k\}$ is **orthonormal** if the vectors are mutually orthogonal (Definition 0.1) and each has norm 1 (Definition 0.2), that is, unit length:

$$\underset{\sim}{x}_i \cdot \underset{\sim}{x}_j = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

Example 0.5 (Indicator vectors are orthonormal)**.** The indicator vectors $\underset{\sim}{e}_1, \underset{\sim}{e}_2, \ldots, \underset{\sim}{e}_p$ (Definition 0.5) form an orthonormal set. By Theorem 0.2, $\underset{\sim}{e}_i \cdot \underset{\sim}{e}_j$ is entry i of $\underset{\sim}{e}_j$, which is 1 if $i = j$ and 0 if $i \neq j$. For $p = 2$:

$$\begin{aligned} \underset{\sim}{e}_1 \cdot \underset{\sim}{e}_1 &= 1 \cdot 1 + 0 \cdot 0 \\ &= 1, \end{aligned}$$

$$\begin{aligned} \underset{\sim}{e}_2 \cdot \underset{\sim}{e}_2 &= 0 \cdot 0 + 1 \cdot 1 \\ &= 1, \end{aligned}$$

and

$$\begin{aligned} \underset{\sim}{e}_1 \cdot \underset{\sim}{e}_2 &= 1 \cdot 0 + 0 \cdot 1 \\ &= 0. \end{aligned}$$

Example 0.6 (Two ways to fail to be orthonormal)**.**

- $(1, 1)$ and $(1, -1)$ are orthogonal, since $1 \cdot 1 + 1 \cdot (-1) = 0$, but not orthonormal: $(1, 1) \cdot (1, 1) = 2 \neq 1$.
- $(1, 0)$ and $(0.6, 0.8)$ both have unit length, since $(1, 0) \cdot (1, 0) = 1$ and

  $$\begin{aligned} (0.6, 0.8) \cdot (0.6, 0.8) &= 0.36 + 0.64 \\ &= 1, \end{aligned}$$

  but they are not orthonormal: $(1, 0) \cdot (0.6, 0.8) = 0.6 \neq 0$.

Exercise 0.5 (Compute an inner product and a norm)**.** Let

$$\underset{\sim}{a} = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}, \qquad \underset{\sim}{b} = \begin{bmatrix} 0 \\ 4 \\ -2 \end{bmatrix}$$

Compute $\underset{\sim}{a}^\top \underset{\sim}{b}$ and $\|\underset{\sim}{a}\|_2$.

Solution

Solution 0.5. Multiply entry by entry and add:

$$\begin{aligned} \underset{\sim}{a} \cdot \underset{\sim}{b} &= (3)(0) + (-1)(4) + (2)(-2) \\ &= 0 - 4 - 4 \\ &= -8 \end{aligned}$$

For the norm, take the inner product of $\underset{\sim}{a}$ with itself first:

$$\begin{aligned} \underset{\sim}{a} \cdot \underset{\sim}{a} &= 3^2 + (-1)^2 + 2^2 \\ &= 9 + 1 + 4 \\ &= 14 \end{aligned}$$

so

$$\|\underset{\sim}{a}\|_2 = \sqrt{14} \approx 3.742$$

The inner product came out negative while the norm cannot: a norm is a square root of a sum of squares, so it is never below zero.
Python has both operations built in, and R builds them from `sum`.

```python
import numpy as np

a = np.array([3.0, -1.0, 2.0])
b = np.array([0.0, 4.0, -2.0])

inner = float(a @ b)
norm_a = round(float(np.linalg.norm(a)), 3)

print(inner, norm_a)
#> -8.0 3.742
```

```r
a <- c(3, -1, 2)
b <- c(0, 4, -2)

print(c(sum(a * b), round(sqrt(sum(a^2)), 3)))
#> [1] -8.000  3.742
```

Both print -8 and 3.742.

Video lecture

Hutchinson's Linear Function Basics (24 min) covers dot products and the Euclidean norm, and goes on to hyperplanes (Definition 27 in Matrices) and level sets [12]. The login for the video site is posted on Canvas.

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